Prerna Arote

Security & Privacy Researcher | Digital Financial Systems | Applied Cryptography (ZKP, MPC)

Madrid, Spain • arote.prerna@gmail.com • +34 657324513

LinkedIn • GitHub • Google Scholar • Website

Summary

Security and privacy researcher (PhD, IISc Bangalore) working on attacks and privacy-preserving protocols for digital financial systems: payments, ledgers, and private trading. Combines attack discovery, formal security analysis (UC framework), and prototype implementation using zero-knowledge proofs, MPC, and commitment schemes. Program Committee member, Financial Cryptography 2027; Topic Editor (Guest Associate Editor), *Frontiers in Blockchain*.

Core Skills

Security expertise	Protocol attack analysis, payment and payment-channel security, double-spending analysis, private trading and settlement, MEV and cross-chain security, threat modelling, UC-framework security proofs
Cryptography	Zero-knowledge proofs (zkSNARKs), MPC, Pedersen commitments, Merkle trees, digital and multi-signatures
Security engineering	Identity and access management, single sign-on (SAML, OAuth, Shibboleth, CAS), Active Directory integration
Languages	Python, Solidity, Go; familiar with Rust and C++
Platforms	Bitcoin, Ethereum, Arbitrum, Polygon, Solana, Sui
Tools	Git, Docker, Kubernetes, Linux, Wireshark, tcpdump

Selected Projects

- **CommTLC: securing HTLCs in payment-channel networks** — Designed and implemented a Pedersen-commitment and signature scheme that detects and punishes Fakey, Griefing and Wormhole attacks; proved security in the UC framework and benchmarked against MAPPCN-OR and EAMHL+ (*IET Blockchain* 2025).
- **Q_Pay: low-latency Bitcoin micropayments** — Built a committee-based protocol for fast payment confirmation without locking user collateral; evaluated on an emulated network of 600 Bitcoin nodes (*SN Computer Science* 2026).
- **Privacy gaps in an MPC-based decentralized dark pool (Renegade)** — Security analysis showing that MPC and ZK proofs ensure correctness but not fairness: selective aborts, probing attacks and the discovery layer leak pre- and post-trade information, and relayer centralization adds liveness risks (under review).
- **SoK: privacy-preserving order matching** — Systematized 22 protocols across submission, discovery, matching and settlement, showing that private matching does not imply end-to-end private trading (under review).
- **Liquidity poisoning attacks on multihop cross-chain arbitrage** — Identified and analyzed an attack class against cross-chain arbitrage paths (IEEE ICBC 2026 workshop).
- **Breaking sandwich MEV in ERC-20 DeFi** — Designed context-aware directional constraints to prevent sandwich attacks (IEEE ICBC 2026 poster).
- **DTH: privacy-preserving certificate traceability** — Certificate storage, verification and revocation for calibration hierarchies using Merkle trees and zkSNARKs (in preparation).
- **SC-ToF: smart-contract task offloading for IoT** — Ethereum and iFogSim implementation; completes 20 MB and 200 MB tasks in 0.8 s and 5.2 s (ICIN 2025).

Experience

Postdoctoral Researcher — IMDEA Networks Institute, Madrid *Dec 2025 – present*

- Led a security analysis of an MPC-based decentralized dark pool (Renegade), uncovering fairness and privacy gaps exploitable in practice.
- Led an SoK on privacy-preserving order matching (22 protocols).
- Designing an end-to-end private trading protocol with confidential yet verifiable on-chain settlement.
- Contributed to a research grant proposal on composable privacy for decentralized dark pools (PI: Dr. L. Kiffer).

- Researched cross-chain arbitrage attacks and sandwich-MEV prevention (two single-author IEEE ICBC 2026 papers).

PhD Researcher — Indian Institute of Science (IISc), Bangalore *Aug 2017 – Jul 2025*

- Designed and implemented protocols against payment-channel attacks (CommTLC) and double-spending in Bitcoin micropayments (Q_Pay, ZCC).
- Built Ethereum-based privacy-preserving systems for industrial IoT (DTH, SC-ToF).
- Contributed to a secure IoT calibration project with the University of Strathclyde.

Software Engineer — DIGITS, Bangalore *Dec 2016 – Jul 2017*

Project Associate — Indian Institute of Science (IISc), Bangalore *Mar 2015 – Mar 2016*

- Designed and deployed a campus-wide identity and access management and single sign-on solution using open-source tools (Gluu, SAML, OAuth, Shibboleth, CAS), integrated with Active Directory.

Assistant Professor, CSE — Symbiosis Institute of Technology, Pune *Jul 2013 – Nov 2014*

Publications

Preprints (under review)

- **P. Arote**, A. Saiz, L. Kiffer. "SoK: Privacy-Preserving Order Matching." arXiv:2610.12535, 2026.
- **P. Arote**, A. Saiz, O. Saguillo, L. Kiffer. "Cryptographic Security Is Not Enough: Privacy Gaps in the Renegade Decentralized Dark Pool." arXiv:2609.27100, 2026.

Journals

- **P. Arote**, J. Kuri. "Q_Pay: A Low-Latency Bitcoin Micropayment Protocol for Instant Service Access." *SN Computer Science* 7(5), 1–37, 2026.
- **P. Arote**, J. Kuri. "CommTLC: An Alternative to Reduce the Attack Surface of HTLCs in Lightning Networks." *IET Blockchain* 5(1), 1–16, 2025.

Conferences and Workshops

- **P. Arote**. "Mind the Gap: Liquidity Poisoning Attacks on Multihop Cross-Chain Arbitrage." *IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, workshop paper, Brisbane, 2026.
- **P. Arote**. "Breaking Sandwich MEV in ERC-20 DeFi via Context-Aware Directional Constraints." *IEEE ICBC*, poster, Brisbane, 2026.
- **P. Arote**, J. Kuri. "SC-ToF: A Secure and Reliable Task Offloading Framework for IoT Devices Using Smart Contracts and Edge Computing." *28th Conference on Innovation in Clouds, Internet and Networks (ICIN)*, Paris, 2025.
- **P. Arote**, J. Kuri. "Detect and Isolate an Adversary in Fakey and Griefing-R Attack on Lightning Network." *7th IEEE International Conference on Blockchain (Blockchain)*, 2024.
- **P. Arote**, J. Kuri. "A Study of Double-spending Causes and Countermeasures in Bitcoin Network." *6th IEEE International Conference on Blockchain Computing and Applications (BCCA)*, 2024.
- **P. Arote**, J. Kuri. "ZCC: Mitigating Double-spending Attacks in Micropayment Bitcoin Transactions." *4th International Conference on Blockchain Computing and Applications (BCCA)*, 2022.
- **P. Arote**, J. Kuri. "Hybrid Decentralized Solution for Bitcoin Zero-confirmation Transactions." *Blockchain Workshop, COMSNETS*, 2019.
- R. Shah, M. McIntee, S. Nagaraja, S. Bhandary, **P. Arote**, J. Kuri. "Secure Calibration in High-Assurance IoT: Traceability for Safety Resilience." arXiv, 2019.
- **P. Arote**, K. V. Arya. "Detection and Prevention against ARP Poisoning Attack Using Modified ICMP and Voting." *International Conference on Computational Intelligence and Networks (CINE)*, 2015.

Education

PhD, Electronic Systems Engineering — Indian Institute of Science (IISc), Bangalore *2017 – 2025*

Thesis: Enhancing Blockchain Security and Efficiency: Solutions for Micropayments, Payment Channels, and IoT Applications.

M.Tech, Computer Science (Information Security) — IIIT Gwalior *2011 – 2013*

Thesis: A Feasible Centralized Framework for the Detection and Prevention of ARP Poisoning Attacks.

Community & Service

- Program committee: Financial Cryptography and Data Security (FC 2027), IEEE BCCA 2025.
- External reviewer: IEEE S&P 2026, AFT 2026, ACM IMC 2026.
- Topic Editor (Guest Associate Editor), *Frontiers in Blockchain*, Research Topic "Security, Economics, and MEV in Decentralized Finance Systems" (2025–present).
- Journal reviewer: Journal of Computer Security (JCS), Blockchain: Research and Applications (BCRA), Computers & Security, Digital Communications and Networks (DCN), Financial Innovation.